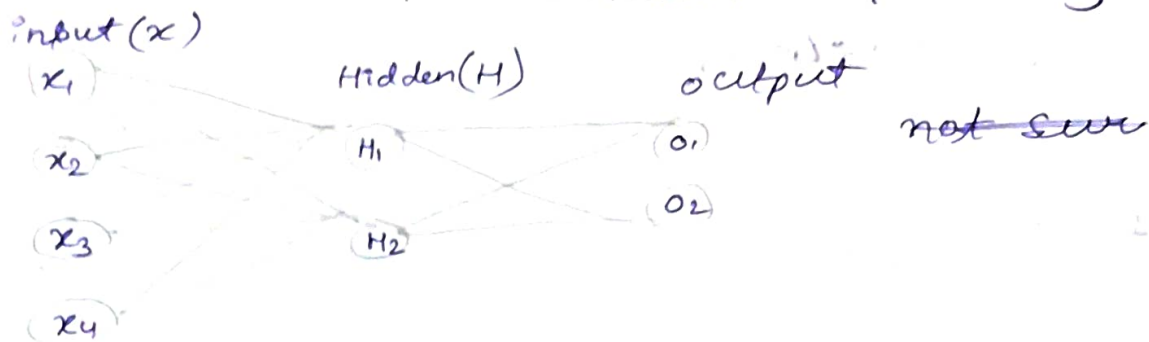


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Class - BCA Sem - 5th

Banknote Authentication Dataset -

The Banknote Authentication Dataset is a classification dataset used to predict whether a banknote is authentic or forged based on statistical features extracted from images of banknotes.

The dataset has 4 input features & 2 output classes.
Input features:- Variance, skewness, curtosis, Img Entropy
output classes $\rightarrow$ 0 $\rightarrow$ Authentic, 1 $\rightarrow$ Forged.



First Row $\rightarrow X = [3.6216, 8.661, -2.8073, -0.44699]$
Actual output = Authentic = 0

Hidden layer $\rightarrow$

For H_1 : $w_1 = 0.01$, $w_2 = 0.02$, $w_3 = 0.01$, $w_4 = 0.03$
 ~~w_5~~ $B_1 = 0.5$

Hidden neuron H_1

$$\begin{aligned} Z_1 &= X_1 w_1 + X_2 w_2 + X_3 w_3 + X_4 w_4 + B_1 \\ &\Rightarrow (3.6216 \times 0.01) + (8.661 \times 0.02) + (-2.8073 \times 0.01) + \\ &\quad (-0.44699 \times 0.03) + 0.5 \\ &\Rightarrow 0.036216 + 0.17322 + 0.5 - 0.0134097 \end{aligned}$$

$$Z_1 = 0.6680553$$

Activation $\rightarrow$

$$\sigma(z_1) = \frac{1}{1 + e^{-z_1}}$$

$$H_1 = \frac{1}{1 + e^{(-0.6680553)}}$$

$$H_1 \approx 0.6611$$

Hidden neuron H_2 +

$$w_5 = 0.02, w_6 = 0.01, w_7 = 0.02, w_8 = 0.01, B_2 = 0.3$$

calculation:

$$Z_2 = X_1 w_5 + X_2 w_6 + X_3 w_7 + X_4 w_8 + B_2$$

$$= (3.6216 \times 0.02) + (8.6661 \times 0.01) + (-2.8073 \times 0.02) + (-0.44699 \times 0.01) + 0.3$$

$$\Rightarrow 0.072432 + 0.086661 - 0.056146 - 0.0044699 + 0.3$$

$$Z_2 = 0.3984771$$

Activation:

$$\sigma(Z_2) = \frac{1}{1 + e^{-(Z_2)}}$$

$$H_2 \cong 0.5983$$

output layer :- O_1 + Authentic + Let:

$$\bullet WH_{1,01} = 0.4$$

$$\bullet WH_{2,01} = 0.2$$

$$\bullet B_{01} = 0.1$$

$$\therefore Z_{\text{Authentic}} = H_1(0.4) + H_2(0.2) + 0.1$$

$$= (0.6611 \times 0.4) + (0.5983 \times 0.2) + 0.1$$

$$Z_{\text{Authentic}} \cong 0.4841$$

O_2 + Forged + Let: $\bullet WH_{1,02} = 0.3, WH_{2,02} = 0.8, B_{02} = 0.1$

$$\therefore Z_{\text{Forged}} = H_1(0.3) + H_2(0.8) + 0.1$$

$$= (0.6611 \times 0.3) + (0.5983 \times 0.8) + 0.1$$

$$Z_{\text{forged}} \cong 0.7770$$

Output value + $[0.4841, 0.7770]$

$$\therefore 0.7770 > 0.4841$$

The highest value is forged.

Predicted output = forged (+)